

Aim: To perform data analysis using Microsoft Excel.**Theory:****Q1. What is Microsoft Excel?**

Microsoft Excel is a powerful spreadsheet application developed by Microsoft. It is used for organizing, storing, calculating, analyzing, and visualizing data. Excel provides features like formulas, functions, charts, pivot tables, sorting, filtering, and conditional formatting to manage data efficiently.

Q2. What is data analysis?

Data analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making. In Excel, data analysis involves using tools like sorting, filtering, pivot tables, charts, and formulas to identify patterns, trends, and insights from raw data.

Q3. What is the purpose of sorting and filtering?

- Sorting arranges data in a specific order (ascending or descending) based on one or more columns. This helps in organizing data alphabetically, numerically, or by date for easier viewing and comparison.
- Filtering displays only the rows that meet certain conditions while hiding the rest. This helps in focusing on specific subsets of data, such as showing only students who scored above 80 marks or sales in a particular region.

Q4. What is a Pivot Table?

A Pivot Table is an interactive Excel tool that summarizes and analyzes large datasets dynamically. It allows users to reorganize, group, aggregate (sum, count, average), and compare data without changing the original dataset. Pivot Tables are useful for creating quick reports and identifying patterns from complex data.

Q5. Name any two Excel functions and their use.

- LEN() – Counts the number of characters in a cell. For example, `=LEN("Hello")` returns 5. It is useful for validating data length, such as checking if a phone number or ID has the correct number of digits.
- IF() – Performs a logical test and returns one value if the condition is true and another value if false. For example, `=IF(A1>75, "Pass", "Fail")` returns "Pass" if the value in cell A1 is greater than 75, otherwise "Fail". It is useful for grading, categorizing data, or creating conditional outputs.

Q6. What is conditional formatting?

Conditional formatting is an Excel feature that automatically applies formatting (such as colors, icons, or data bars) to cells based on specific conditions or rules. For example, you can highlight all cells with values above 80 in green or below 35 in red. It helps in visually identifying trends, outliers, or important data points at a glance.

Q7. Perform data analysis using Excel on a dataset.

Open Excel and load a dataset.

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
101	Aarav Sharma	Mathematics	92	95	A	Excellent
102	Vihaan Gupta	Mathematics	78	82	B	Good
103	Ananya Verma	Mathematics	65	70	C	Average
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
105	Arjun Singh	Mathematics	88	90	A	Very Good
106	Sara Khan	Science	95	98	A	Excellent
107	Rohan Mehta	Science	72	75	B	Good
108	Kriti Joshi	Science	58	65	C	Average
109	Aditya Patil	Science	82	85	B+	Good
110	Neha Reddy	Science	38	50	E	Needs Improvement

Perform sorting (ascending/descending).

Ascending

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
110	Neha Reddy	Science	38	50	E	Needs Improvement
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
108	Kriti Joshi	Science	58	65	C	Average
103	Ananya Verma	Mathematics	65	70	C	Average
107	Rohan Mehta	Science	72	75	B	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
109	Aditya Patil	Science	82	85	B+	Good
105	Arjun Singh	Mathematics	88	90	A	Very Good
101	Aarav Sharma	Mathematics	92	95	A	Excellent
106	Sara Khan	Science	95	98	A	Excellent

Descending

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
106	Sara Khan	Science	95	98	A	Excellent
101	Aarav Sharma	Mathematics	92	95	A	Excellent
105	Arjun Singh	Mathematics	88	90	A	Very Good
109	Aditya Patil	Science	82	85	B+	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
107	Rohan Mehta	Science	72	75	B	Good
103	Ananya Verma	Mathematics	65	70	C	Average
108	Kriti Joshi	Science	58	65	C	Average
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
110	Neha Reddy	Science	38	50	E	Needs Improvement

Apply filters to the dataset.

Filter by only Science

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
106	Sara Khan	Science	95	98	A	Excellent
109	Aditya Patil	Science	82	85	B+	Good
107	Rohan Mehta	Science	72	75	B	Good
108	Kriti Joshi	Science	58	65	C	Average
110	Neha Reddy	Science	38	50	E	Needs Improvement

Filter by only grade A

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
106	Sara Khan	Science	95	98	A	Excellent
101	Aarav Sharma	Mathematics	92	95	A	Excellent
105	Arjun Singh	Mathematics	88	90	A	Very Good

Use conditional formatting (Greater than / Less than).

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
106	Sara Khan	Science	95	98	A	Excellent
101	Aarav Sharma	Mathematics	92	95	A	Excellent
105	Arjun Singh	Mathematics	88	90	A	Very Good
109	Aditya Patil	Science	82	85	B+	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
107	Rohan Mehta	Science	72	75	B	Good
103	Ananya Verma	Mathematics	65	70	C	Average
108	Kriti Joshi	Science	58	65	C	Average
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
110	Neha Reddy	Science	38	50	E	Needs Improvement

Apply Data Bars.

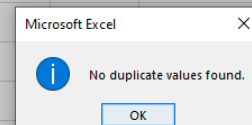
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101	Aarav Sharma	Mathematics	92	95	A	Excellent
105	Arjun Singh	Mathematics	88	90	A	Very Good
109	Aditya Patil	Science	82	85	B+	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
107	Rohan Mehta	Science	72	75	B	Good
103	Ananya Verma	Mathematics	65	70	C	Average
108	Kriti Joshi	Science	58	65	C	Average
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
110	Neha Reddy	Science	38	50	E	Needs Improvement

Apply Color Scales.

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
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105	Arjun Singh	Mathematics	88	90	A	Very Good
109	Aditya Patil	Science	82	85	B+	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
107	Rohan Mehta	Science	72	75	B	Good
103	Ananya Verma	Mathematics	65	70	C	Average
108	Kriti Joshi	Science	58	65	C	Average
104	Diya Nair	Mathematics	45	60	D	Needs Improvement
110	Neha Reddy	Science	38	50	E	Needs Improvement

Remove duplicate values.

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks
106	Sara Khan	Science	95	98	A	Excellent
101	Aarav Sharma	Mathematics	92	95	A	Excellent
105	Arjun Singh	Mathematics	88	90	A	Very Good
109	Aditya Patil	Science	82	85	B+	Good
102	Vihaan Gupta	Mathematics	78	82	B	Good
107	Rohan Mehta	Science	72	75	B	Good
103	Ananya Verma	Mathematics	65			
108	Kriti Joshi	Science	58			
104	Diya Nair	Mathematics	45			Improvement
110	Neha Reddy	Science	38	50	E	Needs Improvement



Create a Pivot Table for analysis.

Row Labels	Average of Marks (Out of 100)
Mathematics	73.6
Science	69
Grand Total	71.3

Average of student marks

Row Labels	Count of Student Name
A	3
B	2
B+	1
C	2
D	1
E	1
Grand Total	10

Grade of students summed

Use functions like IF(), LEN(), or CONCATENATE().

=IF(D2>=50, "Pass", "Fail")

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks	
106	Sara Khan	Science	95	98	A	Excellent	Pass
101	Aarav Sharma	Mathematics	92	95	A	Excellent	Pass
105	Arjun Singh	Mathematics	88	90	A	Very Good	Pass
109	Aditya Patil	Science	82	85	B+	Good	Pass
102	Vihaan Gupta	Mathematics	78	82	B	Good	Pass
107	Rohan Mehta	Science	72	75	B	Good	Pass
103	Ananya Verma	Mathematics	65	70	C	Average	Pass
108	Kriti Joshi	Science	58	65	C	Average	Pass
104	Diya Nair	Mathematics	45	60	D	Needs Improvement	Fail
110	Neha Reddy	Science	38	50	E	Needs Improvement	Fail

=LEN(B2)

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks		
106	Sara Khan	Science	95	98	A	Excellent	Pass	9
101	Aarav Sharma	Mathematics	92	95	A	Excellent	Pass	12
105	Arjun Singh	Mathematics	88	90	A	Very Good	Pass	11
109	Aditya Patil	Science	82	85	B+	Good	Pass	12
102	Vihaan Gupta	Mathematics	78	82	B	Good	Pass	12
107	Rohan Mehta	Science	72	75	B	Good	Pass	11
103	Ananya Verma	Mathematics	65	70	C	Average	Pass	12
108	Kriti Joshi	Science	58	65	C	Average	Pass	11
104	Diya Nair	Mathematics	45	60	D	Needs Improvement	Fail	9
110	Neha Reddy	Science	38	50	E	Needs Improvement	Fail	10

=CONCATENATE(B2, " - ", F2)

StudentID	Student Name	Subject	Marks (Out of 100)	Attendance (%)	Grade	Remarks		
106	Sara Khan	Science	95	98	A	Excellent	Pass	9 Sara Khan - A
101	Aarav Sharma	Mathematics	92	95	A	Excellent	Pass	12 Aarav Sharma - A
105	Arjun Singh	Mathematics	88	90	A	Very Good	Pass	11 Arjun Singh - A
109	Aditya Patil	Science	82	85	B+	Good	Pass	12 Aditya Patil - B+
102	Vihaan Gupta	Mathematics	78	82	B	Good	Pass	12 Vihaan Gupta - B
107	Rohan Mehta	Science	72	75	B	Good	Pass	11 Rohan Mehta - B
103	Ananya Verma	Mathematics	65	70	C	Average	Pass	12 Ananya Verma - C
108	Kriti Joshi	Science	58	65	C	Average	Pass	11 Kriti Joshi - C
104	Diya Nair	Mathematics	45	60	D	Needs Improvement	Fail	9 Diya Nair - D
110	Neha Reddy	Science	38	50	E	Needs Improvement	Fail	10 Neha Reddy - E

Conclusion:

Successfully performed data analysis using Microsoft Excel.